

Lab Report 5 - Project 1 - Cybersecurity Analytics

Use Different Tools to Analyze Data from a Cybersecurity Incident

Introduction

This lab is designed to work with VM machines to get practical skills of analyzing captured traffic from the *file_lab2.pcap* file using Wireshark, Network Miner and *lab5_procmon.PML* file using Process Monitor to investigate lateral movement and data exfiltration occurred on victim IP address 10.0.2.4.

Using Wireshark tool, we can analyze the capture packet to detect protocol sessions and inspect commands exchanged between the client and the server. The Network Miner can be used to investigate communication flow and extract host details, session logs, transfer file details, and credentials. The ProcMon filters can be used to identify TCP connect, send, and receive operations and system level confirmation of and transfer files through specific ephemeral ports. It is required to continuously monitor network systems to identify vulnerabilities and implement controls to mitigate involved treats.

This investigation aims to evaluate network activity and potential security breach using network and process monitoring tools. After evaluating the results of investigation will proceed to introduce necessary security control measures and policy to safeguard the XFibility bank network.

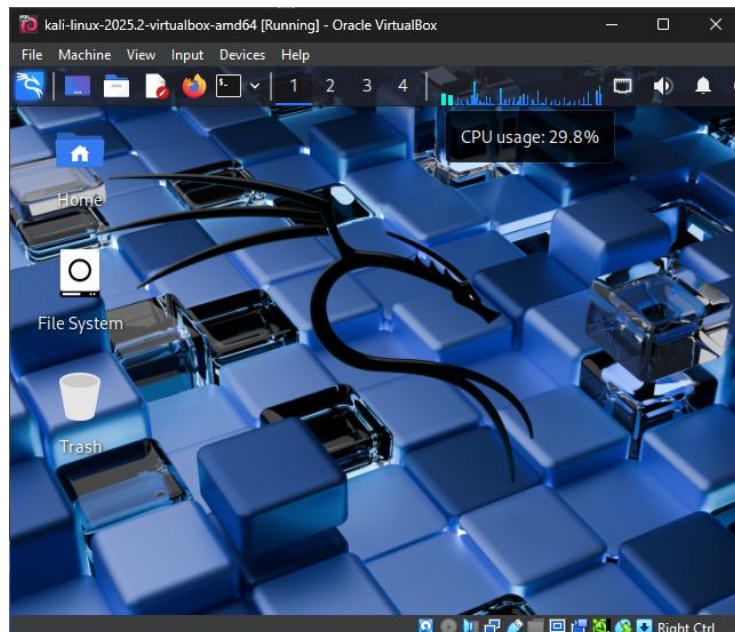
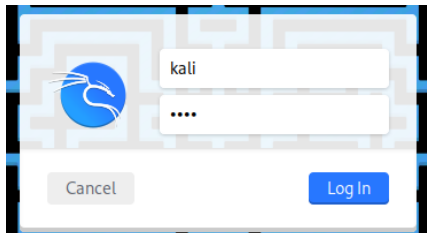
Step 1 – Using Wireshark

1. Logged to Azure Lab environment using following web link and logged into the Kali VM using Oracle VirtualBox.

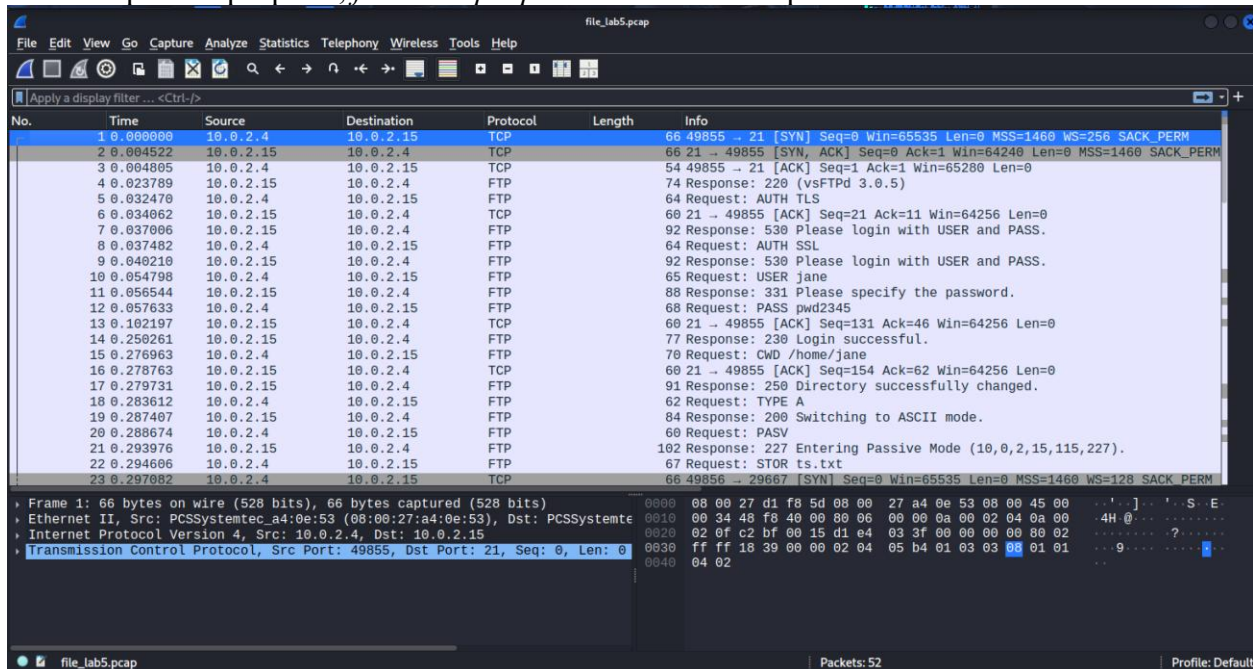
https://labs.azure.com/virtualmachines?feature_vnext=true

Username - kali

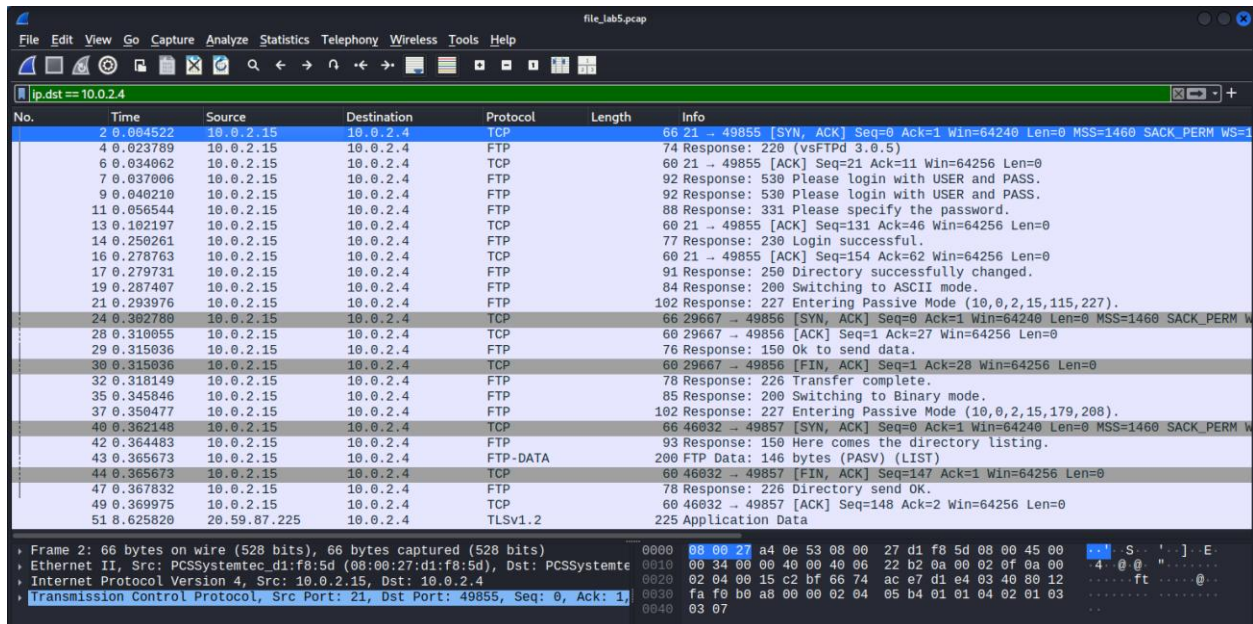
Password - kali



2. Open the pcap file, *file lab5.pcap* on Wireshark to capture the data.



3. Used *ip.dst == 10.0.2.4* display filter to capture the traffic coming to the Victim. I found that 26 packets reached the Victim IP and all came from two source IPs. TCP and FTP protocols were used to reach the destination IP (Victim IP) and successful FTP access shown in the captured data.



4. Used Statistics > Endpoints to search about IPv4 details. I found that Victim IP received 25 packets from the 10.0.2.15 IP and one packet was received from the 20.59.87.225 IP address.

Wireshark · Endpoints · file_lab5.pcap

Endpoint Settings

- ☐ Name resolution
- ☒ Limit to display filter
- Copy
- Map
- Protocol
 - ☐ Bluetooth
 - ☐ BPv7
 - ☐ DCCP
 - ☒ Ethernet
 - ☐ FC
 - ☐ FDDI
 - ☐ IEEE 802.11
- Filter list for specific type

Address	Packets	Bytes	Total Packets	Percent Filtered	Tx Packets	T
10.0.2.4	26	2 kB	52	50.00%	0	
10.0.2.15	25	2 kB	49	51.02%	25	
20.59.87.225	1	225 bytes	3	33.33%	1	22

Close Help

The victim host 10.0.2.4 established TCP connection from ephemeral client ports 49674, 49855, 49856, and 49857 to remote services. The attacker host, 10.0.2.15 accepted FTP control on ports 21 and used ephemeral data ports 29667 and 46032 for FTP data channels. Also, the victim host made an outbound encrypted connection to 20.59.87.225 through port 443 (HTTPS), which is like external relay.

Wireshark · Endpoints · file_lab5.pcap

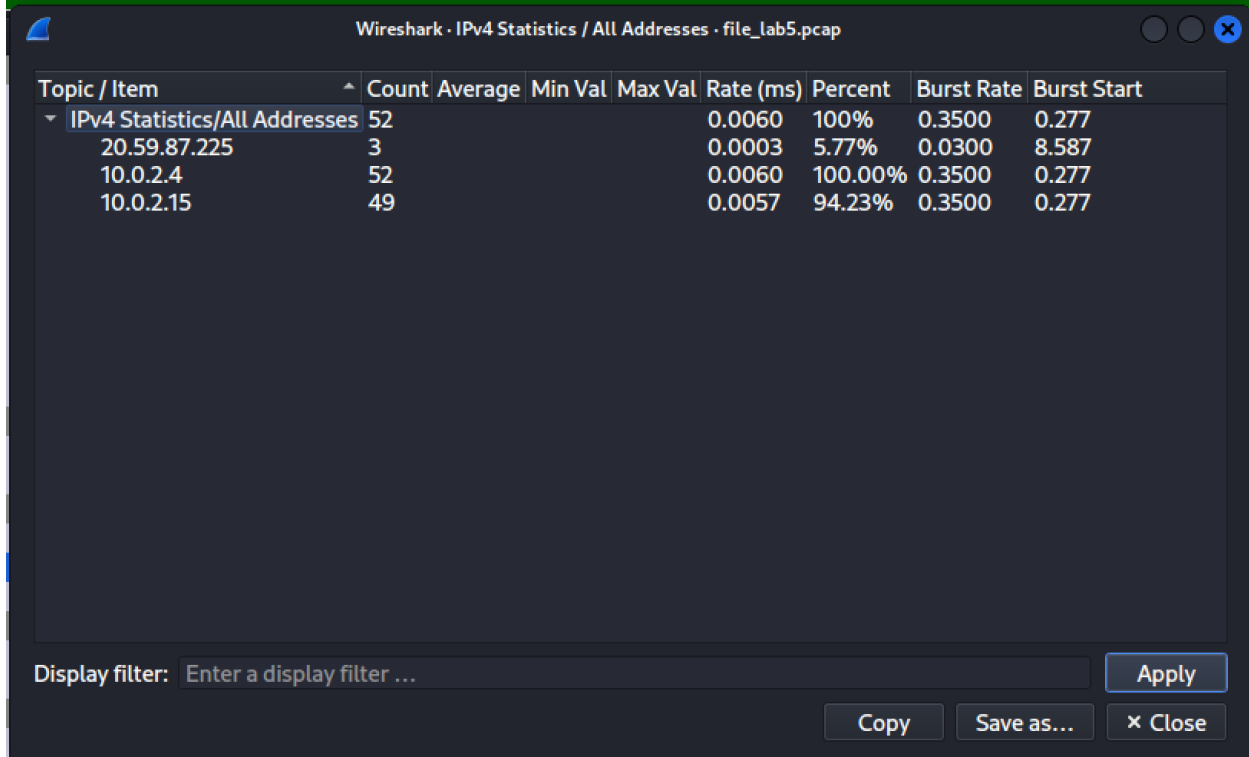
Endpoint Settings

- ☐ Name resolution
- ☒ Limit to display filter
- Copy
- Map
- Protocol
 - ☐ Bluetooth
 - ☐ BPv7
 - ☐ DCCP
 - ☒ Ethernet
 - ☐ FC
 - ☐ FDDI
 - ☐ IEEE 802.11
- Filter list for specific type

Address	Port	Packets	Bytes	Total Packets	Percent Filtered	Tx Pack
10.0.2.4	49674	1	225 bytes	3	33.33%	
10.0.2.4	49855	18	1 kB	33	54.55%	
10.0.2.4	49856	3	186 bytes	8	37.50%	
10.0.2.4	49857	4	386 bytes	8	50.00%	
10.0.2.15	21	18	1 kB	33	54.55%	
10.0.2.15	29667	3	186 bytes	8	37.50%	
10.0.2.15	46032	4	386 bytes	8	50.00%	
20.59.87.225	443	1	225 bytes	3	33.33%	

Close Help

5. Used Statistics > IPv4 Statics > All Addresses and Destinations and Ports to search about IPv4 details. Found the victim IP and two IP used for the attack.



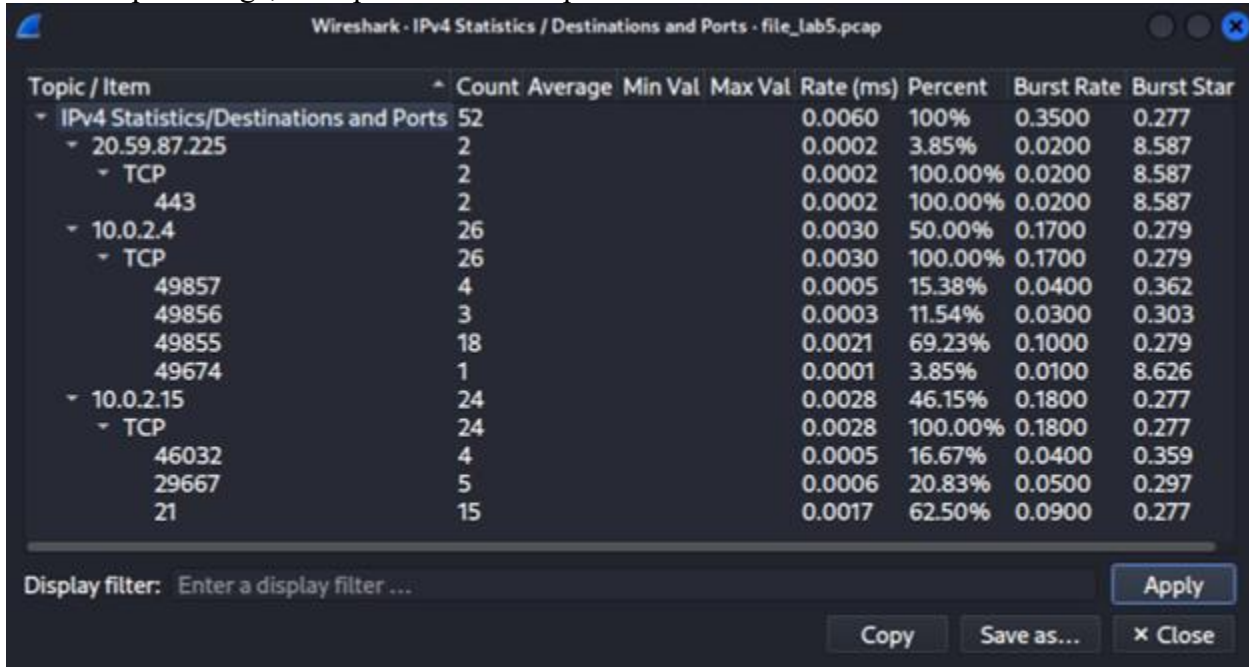
Wireshark - IPv4 Statistics / All Addresses - file_lab5.pcap

Topic / Item	Count	Average	Min Val	Max Val	Rate (ms)	Percent	Burst Rate	Burst Start
IPv4 Statistics/All Addresses	52				0.0060	100%	0.3500	0.277
20.59.87.225	3				0.0003	5.77%	0.0300	8.587
10.0.2.4	52				0.0060	100.00%	0.3500	0.277
10.0.2.15	49				0.0057	94.23%	0.3500	0.277

Display filter: Enter a display filter ...

Copy Save as... Close

Same ports details found on the destinations and ports section in statistics. Also found the details of percentage, each packet count on ports and relevant details.



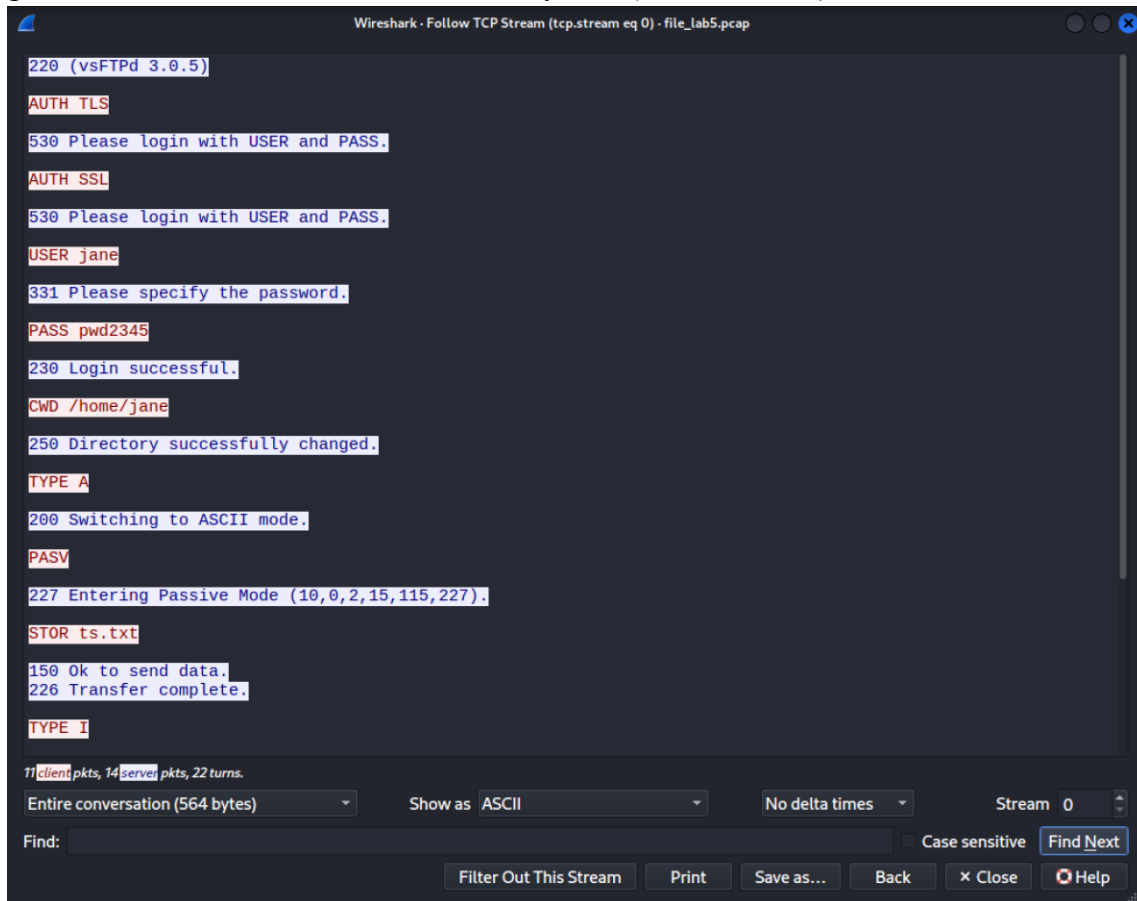
Wireshark - IPv4 Statistics / Destinations and Ports - file_lab5.pcap

Topic / Item	Count	Average	Min Val	Max Val	Rate (ms)	Percent	Burst Rate	Burst Star
IPv4 Statistics/Destinations and Ports	52				0.0060	100%	0.3500	0.277
20.59.87.225	2				0.0002	3.85%	0.0200	8.587
TCP	2				0.0002	100.00%	0.0200	8.587
443	2				0.0002	100.00%	0.0200	8.587
10.0.2.4	26				0.0030	50.00%	0.1700	0.279
TCP	26				0.0030	100.00%	0.1700	0.279
49857	4				0.0005	15.38%	0.0400	0.362
49856	3				0.0003	11.54%	0.0300	0.303
49855	18				0.0021	69.23%	0.1000	0.279
49674	1				0.0001	3.85%	0.0100	8.626
10.0.2.15	24				0.0028	46.15%	0.1800	0.277
TCP	24				0.0028	100.00%	0.1800	0.277
46032	4				0.0005	16.67%	0.0400	0.359
29667	5				0.0006	20.83%	0.0500	0.297
21	15				0.0017	62.50%	0.0900	0.277

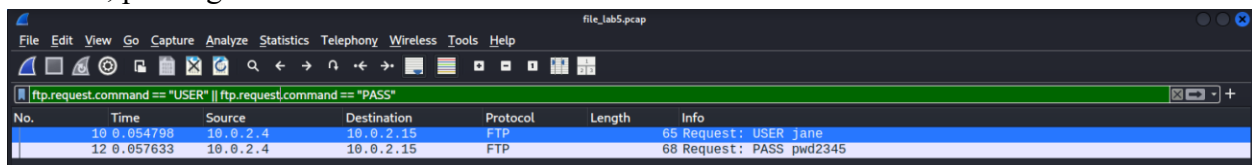
Display filter: Enter a display filter ...

Copy Save as... Close

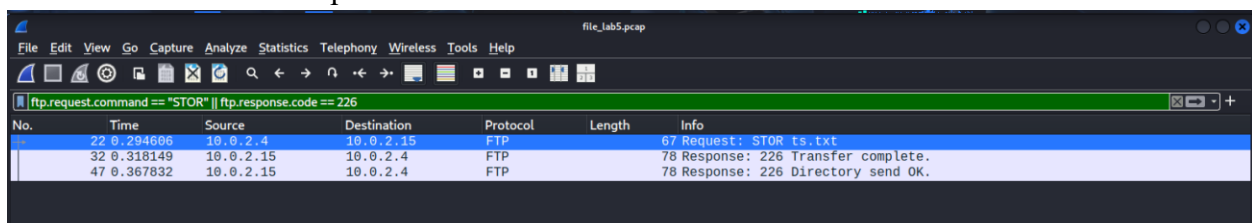
6. Used *ftp* display filter to follow up the session, and then Right clicked a packet and selected Follow > TCP Stream to see entire command exchange. Further found the details of login credentials in clear text (USER jane), and password (PAS pwd2345). This indicated the attacker gained valid authentication to the victim system (Bank network).



7. Used *ftp.request.command == "STOR" || ftp.request.command == "PASS"* filter command to check FTP Credentials. This confirms that the file *ts.txt* was uploaded from the victim to the attacker, proving successful data exfiltration.

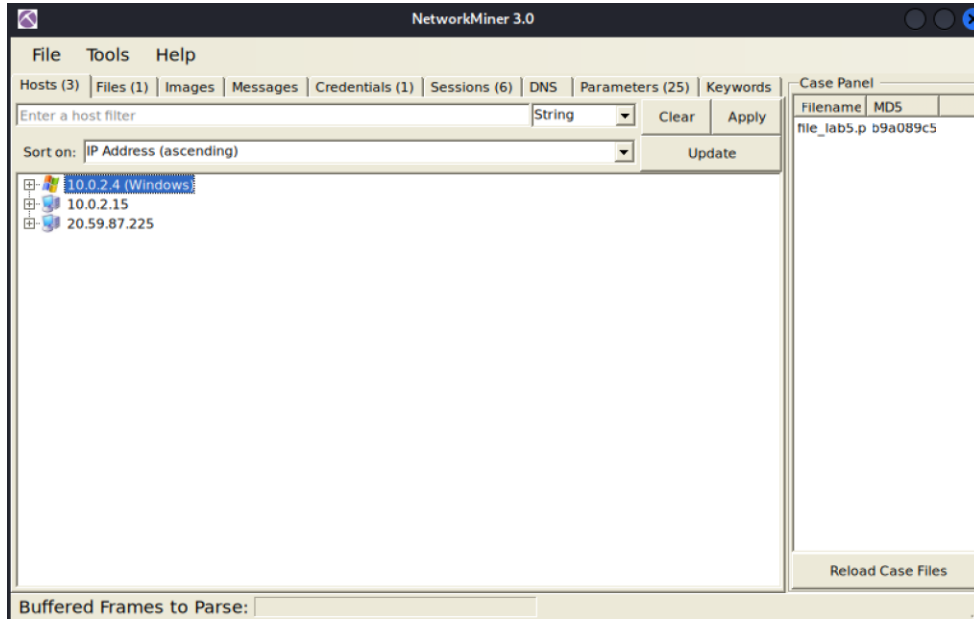


8. Used *ftp.request.command == "STOR" || ftp.response.code == 226* filter command to check file transfers. This proves that file exfiltration from the victim host.

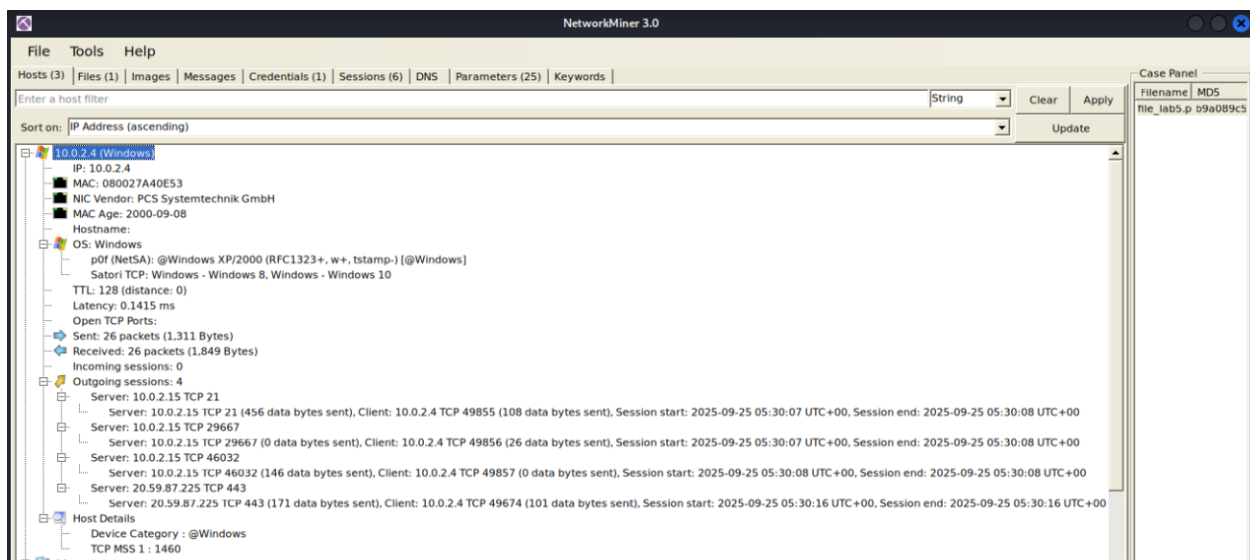


Step 2 – Using Network Miner

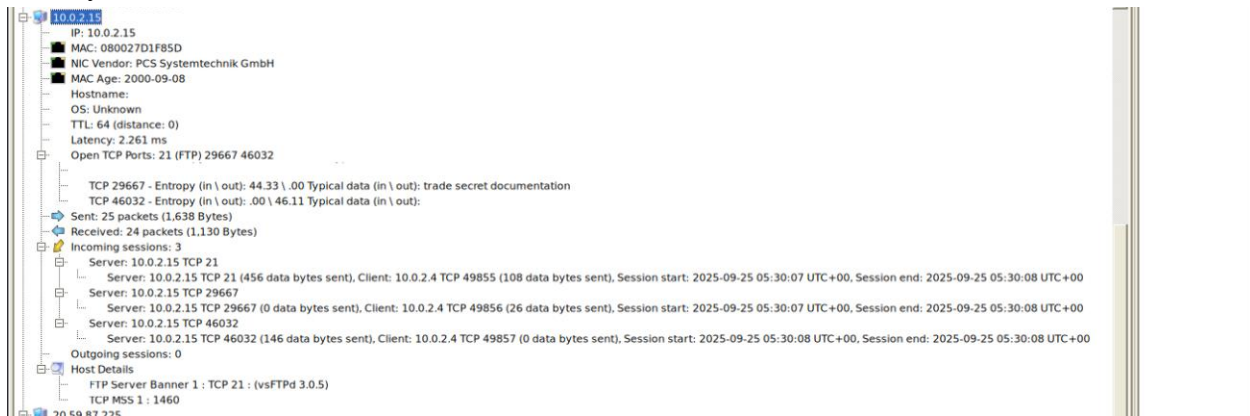
1. Loaded the provided pcap, *file_lab5.pcap* into Network Miner. In the Network Miner Hosts tab, confirmed the same three IPv4 addresses were detected. 10.0.2.4 is victim (internal host) likely workstation and 10.0.2.15 and 20.59.87.225 external hosts.



2. Check the detailed view of victim host 10.0.2.4 and I found the OS running on the host is Windows and sent 26 packets and received 26 packets. Outgoing sessions reveal FTP connections to attacker 10.0.2.15 and encrypted TLS traffic to external host 20.59.87.225 and each port used to establish a connection.



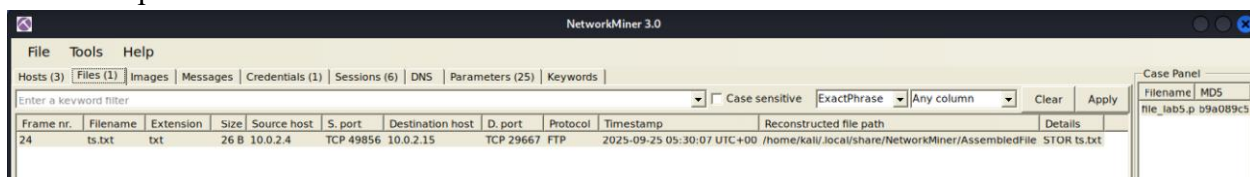
Check the detailed view of attack host 10.0.2.15 and I found the attacker hosting FTP service (port 21) with data channels 29667 and 46032. Also, it includes victim login, file transfer activity, and FTP server banner vsFTPD 3.0.5.



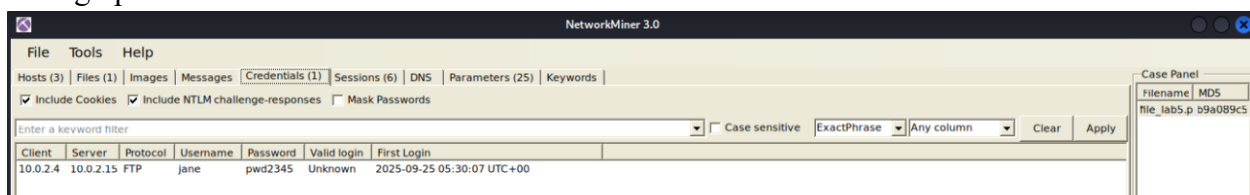
Check the detailed view of external server 20.59.87.225 communicating with victim host 10.0.2.4 over TCP port 443 (HTTPS). TLS session indicated possible forwarding or staging of exfiltrated data on an outside event.



3. Files capture on Network Miner shows the file ts.txt (26 B) transferred from victim host 10.0.2.4 to the attacker host 10.0.2.15 using FTP protocol. Also, it gives details of timestamp and saved file path as forensic evidence.



4. Credentials found on Network Miner show the cleartext FTP credentials, (USER jane), and password (PAS pwd2345). Session shows login from victim host 10.0.2.4 to attacker host 10.0.2.15 through port 21. It confirms that the attacker obtained valid authentication for file transfer.



5. Sessions found on Network Miner listed out six captured sessions. FTP control port 49855 to port 21, and other data channels through ephemeral ports 49856 to port 29667, and 49857 to port 49032. Also, SSL control port 49674 to port 443. The attacker transfers files via FTP and contacted external server.

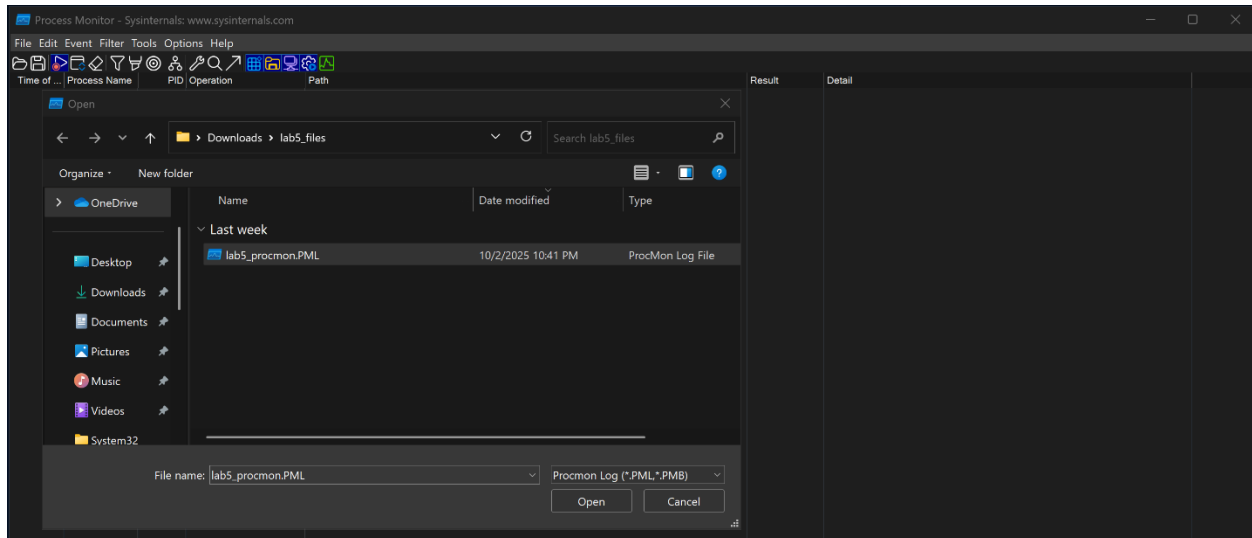
Frame nr.	Client host	C. port	Server host	S. port	Protocol (application layer)	Start time
1	10.0.2.4	49855	10.0.2.15	21	FTP	2025-09-25 05:30:07 UTC+00
23	10.0.2.4	49856	10.0.2.15	29667		2025-09-25 05:30:07 UTC+00
39	10.0.2.4	49857	10.0.2.15	46032		2025-09-25 05:30:08 UTC+00
50	10.0.2.4	49674	20.59.87.225	443	SSL	2025-09-25 05:30:16 UTC+00
39	10.0.2.4	49857	10.0.2.15	46032		2025-09-25 05:30:08 UTC+00
23	10.0.2.4	49856	10.0.2.15	29667		2025-09-25 05:30:07 UTC+00

6. Parameters found on Network Miner shows FTP commands including USER jane, PASS pwd2345, directory changed to /home/jane, and transfer file ts.txt. Responses confirm login successful, directory change, switching mode, and file transfer completion code 226.

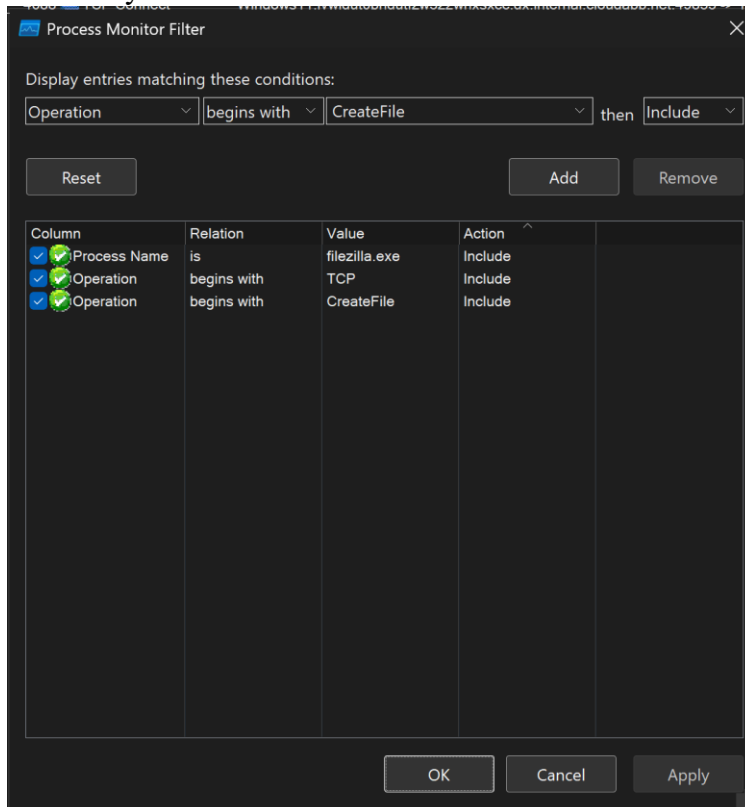
Parameter name	Parameter value	Frame number	Source host	Source port	Destination host	Destination port	Timestamp	Details
220	(vsFTPD 3.0.5)	4	10.0.2.15	TCP 21	10.0.2.4	TCP 49855	2025-09-25 05:30:07 UTC+00	FTP Response
AUTH	TLS	5	10.0.2.4	TCP 49855	10.0.2.15	TCP 21	2025-09-25 05:30:07 UTC+00	FTP Request
530	Please login with USER and PASS.	7	10.0.2.15	TCP 21	10.0.2.4	TCP 49855	2025-09-25 05:30:07 UTC+00	FTP Response
AUTH	SSL	8	10.0.2.4	TCP 49855	10.0.2.15	TCP 21	2025-09-25 05:30:07 UTC+00	FTP Request
530	Please login with USER and PASS.	9	10.0.2.15	TCP 21	10.0.2.4	TCP 49855	2025-09-25 05:30:07 UTC+00	FTP Response
USER	jane	10	10.0.2.4	TCP 49855	10.0.2.15	TCP 21	2025-09-25 05:30:07 UTC+00	FTP Request
331	Please specify the password.	11	10.0.2.15	TCP 21	10.0.2.4	TCP 49855	2025-09-25 05:30:07 UTC+00	FTP Response
PASS	pwd2345	12	10.0.2.4	TCP 49855	10.0.2.15	TCP 21	2025-09-25 05:30:07 UTC+00	FTP Request
230	Login successful.	14	10.0.2.15	TCP 21	10.0.2.4	TCP 49855	2025-09-25 05:30:07 UTC+00	FTP Response
CWD	/home/jane	15	10.0.2.4	TCP 49855	10.0.2.15	TCP 21	2025-09-25 05:30:07 UTC+00	FTP Request
250	Directory successfully changed.	17	10.0.2.15	TCP 21	10.0.2.4	TCP 49855	2025-09-25 05:30:07 UTC+00	FTP Response
TYPE	A	18	10.0.2.4	TCP 49855	10.0.2.15	TCP 21	2025-09-25 05:30:07 UTC+00	FTP Request
200	Switching to ASCII mode.	19	10.0.2.15	TCP 21	10.0.2.4	TCP 49855	2025-09-25 05:30:07 UTC+00	FTP Response
PASV	Switching to ASCII mode.	20	10.0.2.4	TCP 49855	10.0.2.15	TCP 21	2025-09-25 05:30:07 UTC+00	FTP Request
227	Entering Passive Mode (10.0.2.15.115.227).	21	10.0.2.15	TCP 21	10.0.2.4	TCP 49855	2025-09-25 05:30:07 UTC+00	FTP Response
STOR	ts.txt	22	10.0.2.4	TCP 49855	10.0.2.15	TCP 21	2025-09-25 05:30:07 UTC+00	FTP Request
150	Ok to send data.	29	10.0.2.15	TCP 21	10.0.2.4	TCP 49855	2025-09-25 05:30:08 UTC+00	FTP Response
226	Transfer complete.	32	10.0.2.15	TCP 21	10.0.2.4	TCP 49855	2025-09-25 05:30:08 UTC+00	FTP Response
TYPE	I	34	10.0.2.4	TCP 49855	10.0.2.15	TCP 21	2025-09-25 05:30:08 UTC+00	FTP Request
200	Switching to Binary mode.	35	10.0.2.15	TCP 21	10.0.2.4	TCP 49855	2025-09-25 05:30:08 UTC+00	FTP Response
PASV	Switching to Binary mode.	36	10.0.2.4	TCP 49855	10.0.2.15	TCP 21	2025-09-25 05:30:08 UTC+00	FTP Request
227	Entering Passive Mode (10.0.2.15.179.208).	37	10.0.2.15	TCP 21	10.0.2.4	TCP 49855	2025-09-25 05:30:08 UTC+00	FTP Response
LIST		38	10.0.2.4	TCP 49855	10.0.2.15	TCP 21	2025-09-25 05:30:08 UTC+00	FTP Request
150	Here comes the directory listing.	42	10.0.2.15	TCP 21	10.0.2.4	TCP 49855	2025-09-25 05:30:08 UTC+00	FTP Response
226	Directory send OK.	47	10.0.2.15	TCP 21	10.0.2.4	TCP 49855	2025-09-25 05:30:08 UTC+00	FTP Response

Step 3 – Using Process Monitor (ProcMon)

1. Logged in to Windows 11 VM and opened ProcMon and loaded *lab5_procmon.PML* file into the ProcMon.



2. Include rules for filezilla.exe, Operation begins with TCP, and Operation begins with CreateFile filters to monitor behavior of the scenario and isolates FileZilla's network and file access behaviors. This will summarize field we require to focus on getting valuable details related to security incidents.



After filtering out details, I obtained details of FileZilla TCP Connect, TCP Receive, TCP Send, and TCP Disconnect events to 10.0.2.15 and data ports 29667, 46032. After reviewing these results, confirm process activities matching Wireshark FTP data streams and exfiltration.

Time of...	Process Name	PID	Operation	Path	Result	Detail
12:30:07...	filezilla.exe	4688	TCP Connect	Windows11.ipv4to0b9hufzw32wmxsxc:gx.internal.cloudapp.net:49855 -> 10.0.2.15:ftp	SUCCESS	Length: 0, mss: 1460, sackopt: 1, tsopt: 0, wsopt: 1, rcwin: 65535, rcwinscale: 8
12:30:07...	filezilla.exe	4688	TCP TCPCopy	Windows11.ipv4to0b9hufzw32wmxsxc:gx.internal.cloudapp.net:49855 -> 10.0.2.15:ftp	SUCCESS	Length: 20, sequum: 0, connid: 0
12:30:07...	filezilla.exe	4688	TCP Receive	Windows11.ipv4to0b9hufzw32wmxsxc:gx.internal.cloudapp.net:49855 -> 10.0.2.15:ftp	SUCCESS	Length: 20, sequum: 0, connid: 0
12:30:07...	filezilla.exe	4688	TCP Send	Windows11.ipv4to0b9hufzw32wmxsxc:gx.internal.cloudapp.net:49855 -> 10.0.2.15:ftp	SUCCESS	Length: 10, starttime: 203484, endtime: 203485, sequum: 0, connid: 0
12:30:07...	filezilla.exe	4688	TCP TCPCopy	Windows11.ipv4to0b9hufzw32wmxsxc:gx.internal.cloudapp.net:49855 -> 10.0.2.15:ftp	SUCCESS	Length: 38, sequum: 0, connid: 0
12:30:07...	filezilla.exe	4688	TCP Receive	Windows11.ipv4to0b9hufzw32wmxsxc:gx.internal.cloudapp.net:49855 -> 10.0.2.15:ftp	SUCCESS	Length: 38, sequum: 0, connid: 0
12:30:07...	filezilla.exe	4688	TCP Send	Windows11.ipv4to0b9hufzw32wmxsxc:gx.internal.cloudapp.net:49855 -> 10.0.2.15:ftp	SUCCESS	Length: 10, starttime: 203485, endtime: 203485, sequum: 0, connid: 0
12:30:07...	filezilla.exe	4688	TCP TCPCopy	Windows11.ipv4to0b9hufzw32wmxsxc:gx.internal.cloudapp.net:49855 -> 10.0.2.15:ftp	SUCCESS	Length: 38, sequum: 0, connid: 0
12:30:07...	filezilla.exe	4688	TCP Receive	Windows11.ipv4to0b9hufzw32wmxsxc:gx.internal.cloudapp.net:49855 -> 10.0.2.15:ftp	SUCCESS	Length: 38, sequum: 0, connid: 0
12:30:07...	filezilla.exe	4688	TCP Send	Windows11.ipv4to0b9hufzw32wmxsxc:gx.internal.cloudapp.net:49855 -> 10.0.2.15:ftp	SUCCESS	Length: 11, starttime: 203487, endtime: 203487, sequum: 0, connid: 0
12:30:07...	filezilla.exe	4688	TCP TCPCopy	Windows11.ipv4to0b9hufzw32wmxsxc:gx.internal.cloudapp.net:49855 -> 10.0.2.15:ftp	SUCCESS	Length: 34, sequum: 0, connid: 0
12:30:07...	filezilla.exe	4688	TCP Receive	Windows11.ipv4to0b9hufzw32wmxsxc:gx.internal.cloudapp.net:49855 -> 10.0.2.15:ftp	SUCCESS	Length: 34, sequum: 0, connid: 0
12:30:07...	filezilla.exe	4688	TCP Send	Windows11.ipv4to0b9hufzw32wmxsxc:gx.internal.cloudapp.net:49855 -> 10.0.2.15:ftp	SUCCESS	Length: 14, starttime: 203487, endtime: 203491, sequum: 0, connid: 0
12:30:07...	filezilla.exe	4688	TCP TCPCopy	Windows11.ipv4to0b9hufzw32wmxsxc:gx.internal.cloudapp.net:49855 -> 10.0.2.15:ftp	SUCCESS	Length: 23, sequum: 0, connid: 0
12:30:07...	filezilla.exe	4688	TCP Receive	Windows11.ipv4to0b9hufzw32wmxsxc:gx.internal.cloudapp.net:49855 -> 10.0.2.15:ftp	SUCCESS	Length: 23, sequum: 0, connid: 0
12:30:07...	filezilla.exe	4688	TCP Send	Windows11.ipv4to0b9hufzw32wmxsxc:gx.internal.cloudapp.net:49855 -> 10.0.2.15:ftp	SUCCESS	Length: 16, starttime: 203509, endtime: 203509, sequum: 0, connid: 0
12:30:07...	filezilla.exe	4688	TCP TCPCopy	Windows11.ipv4to0b9hufzw32wmxsxc:gx.internal.cloudapp.net:49855 -> 10.0.2.15:ftp	SUCCESS	Length: 37, sequum: 0, connid: 0
12:30:07...	filezilla.exe	4688	TCP Receive	Windows11.ipv4to0b9hufzw32wmxsxc:gx.internal.cloudapp.net:49855 -> 10.0.2.15:ftp	SUCCESS	Length: 37, sequum: 0, connid: 0
12:30:07...	filezilla.exe	4688	TCP Send	Windows11.ipv4to0b9hufzw32wmxsxc:gx.internal.cloudapp.net:49855 -> 10.0.2.15:ftp	SUCCESS	Length: 8, starttime: 203510, endtime: 203510, sequum: 0, connid: 0
12:30:07...	filezilla.exe	4688	TCP TCPCopy	Windows11.ipv4to0b9hufzw32wmxsxc:gx.internal.cloudapp.net:49855 -> 10.0.2.15:ftp	SUCCESS	Length: 30, sequum: 0, connid: 0
12:30:07...	filezilla.exe	4688	TCP Receive	Windows11.ipv4to0b9hufzw32wmxsxc:gx.internal.cloudapp.net:49855 -> 10.0.2.15:ftp	SUCCESS	Length: 30, sequum: 0, connid: 0
12:30:07...	filezilla.exe	4688	TCP Send	Windows11.ipv4to0b9hufzw32wmxsxc:gx.internal.cloudapp.net:49855 -> 10.0.2.15:ftp	SUCCESS	Length: 6, starttime: 203510, endtime: 203511, sequum: 0, connid: 0
12:30:07...	filezilla.exe	4688	TCP TCPCopy	Windows11.ipv4to0b9hufzw32wmxsxc:gx.internal.cloudapp.net:49855 -> 10.0.2.15:ftp	SUCCESS	Length: 48, sequum: 0, connid: 0
12:30:07...	filezilla.exe	4688	TCP Receive	Windows11.ipv4to0b9hufzw32wmxsxc:gx.internal.cloudapp.net:49855 -> 10.0.2.15:ftp	SUCCESS	Length: 48, sequum: 0, connid: 0
12:30:07...	filezilla.exe	4688	TCP Connect	Windows11.ipv4to0b9hufzw32wmxsxc:gx.internal.cloudapp.net:49856 -> 10.0.2.15:29667	SUCCESS	Length: 0, mss: 1460, sackopt: 1, tsopt: 0, wsopt: 1, rcwin: 4194304, rcwinscale:...
12:30:07...	filezilla.exe	4688	TCP Send	Windows11.ipv4to0b9hufzw32wmxsxc:gx.internal.cloudapp.net:49856 -> 10.0.2.15:29667	SUCCESS	Length: 13, starttime: 203511, endtime: 203512, sequum: 0, connid: 0
12:30:08...	filezilla.exe	4688	TCP TCPCopy	Windows11.ipv4to0b9hufzw32wmxsxc:gx.internal.cloudapp.net:49855 -> 10.0.2.15:ftp	SUCCESS	Length: 22, sequum: 0, connid: 0
12:30:08...	filezilla.exe	4688	TCP Receive	Windows11.ipv4to0b9hufzw32wmxsxc:gx.internal.cloudapp.net:49855 -> 10.0.2.15:ftp	SUCCESS	Length: 22, sequum: 0, connid: 0
12:30:08...	filezilla.exe	4688	TCP Disconnect	Windows11.ipv4to0b9hufzw32wmxsxc:gx.internal.cloudapp.net:49856 -> 10.0.2.15:29667	SUCCESS	Length: 0, sequum: 0, connid: 0
12:30:08...	filezilla.exe	4688	TCP TCPCopy	Windows11.ipv4to0b9hufzw32wmxsxc:gx.internal.cloudapp.net:49855 -> 10.0.2.15:ftp	SUCCESS	Length: 24, sequum: 0, connid: 0
12:30:08...	filezilla.exe	4688	TCP Receive	Windows11.ipv4to0b9hufzw32wmxsxc:gx.internal.cloudapp.net:49855 -> 10.0.2.15:ftp	SUCCESS	Length: 24, sequum: 0, connid: 0
12:30:08...	filezilla.exe	4688	TCP Send	Windows11.ipv4to0b9hufzw32wmxsxc:gx.internal.cloudapp.net:49855 -> 10.0.2.15:ftp	SUCCESS	Length: 8, starttime: 203515, endtime: 203516, sequum: 0, connid: 0
12:30:08...	filezilla.exe	4688	TCP TCPCopy	Windows11.ipv4to0b9hufzw32wmxsxc:gx.internal.cloudapp.net:49855 -> 10.0.2.15:ftp	SUCCESS	Length: 31, sequum: 0, connid: 0
12:30:08...	filezilla.exe	4688	TCP Receive	Windows11.ipv4to0b9hufzw32wmxsxc:gx.internal.cloudapp.net:49855 -> 10.0.2.15:ftp	SUCCESS	Length: 31, sequum: 0, connid: 0
12:30:08...	filezilla.exe	4688	TCP Send	Windows11.ipv4to0b9hufzw32wmxsxc:gx.internal.cloudapp.net:49855 -> 10.0.2.15:ftp	SUCCESS	Length: 6, starttime: 203516, endtime: 203516, sequum: 0, connid: 0
12:30:08...	filezilla.exe	4688	TCP TCPCopy	Windows11.ipv4to0b9hufzw32wmxsxc:gx.internal.cloudapp.net:49855 -> 10.0.2.15:ftp	SUCCESS	Length: 48, sequum: 0, connid: 0
12:30:08...	filezilla.exe	4688	TCP Receive	Windows11.ipv4to0b9hufzw32wmxsxc:gx.internal.cloudapp.net:49855 -> 10.0.2.15:ftp	SUCCESS	Length: 48, sequum: 0, connid: 0
12:30:08...	filezilla.exe	4688	TCP Connect	Windows11.ipv4to0b9hufzw32wmxsxc:gx.internal.cloudapp.net:49857 -> 10.0.2.15:46032	SUCCESS	Length: 0, mss: 1460, sackopt: 1, tsopt: 0, wsopt: 1, rcwin: 4194304, rcwinscale:...
12:30:08...	filezilla.exe	4688	TCP Send	Windows11.ipv4to0b9hufzw32wmxsxc:gx.internal.cloudapp.net:49857 -> 10.0.2.15:46032	SUCCESS	Length: 8, starttime: 203516, endtime: 203516, sequum: 0, connid: 0
12:30:08...	filezilla.exe	4688	TCP TCPCopy	Windows11.ipv4to0b9hufzw32wmxsxc:gx.internal.cloudapp.net:49855 -> 10.0.2.15:ftp	SUCCESS	Length: 39, sequum: 0, connid: 0
12:30:08...	filezilla.exe	4688	TCP Receive	Windows11.ipv4to0b9hufzw32wmxsxc:gx.internal.cloudapp.net:49855 -> 10.0.2.15:ftp	SUCCESS	Length: 39, sequum: 0, connid: 0
12:30:08...	filezilla.exe	4688	TCP TCPCopy	Windows11.ipv4to0b9hufzw32wmxsxc:gx.internal.cloudapp.net:49857 -> 10.0.2.15:46032	SUCCESS	Length: 146, sequum: 0, connid: 0
12:30:08...	filezilla.exe	4688	TCP Receive	Windows11.ipv4to0b9hufzw32wmxsxc:gx.internal.cloudapp.net:49857 -> 10.0.2.15:46032	SUCCESS	Length: 146, sequum: 0, connid: 0
12:30:08...	filezilla.exe	4688	TCP Send	Windows11.ipv4to0b9hufzw32wmxsxc:gx.internal.cloudapp.net:49855 -> 10.0.2.15:ftp	SUCCESS	Length: 24, sequum: 0, connid: 0
12:30:08...	filezilla.exe	4688	TCP TCPCopy	Windows11.ipv4to0b9hufzw32wmxsxc:gx.internal.cloudapp.net:49855 -> 10.0.2.15:ftp	SUCCESS	Length: 24, sequum: 0, connid: 0
12:30:08...	filezilla.exe	4688	TCP Disconnect	Windows11.ipv4to0b9hufzw32wmxsxc:gx.internal.cloudapp.net:49857 -> 10.0.2.15:46032	SUCCESS	Length: 0, sequum: 0, connid: 0

3. Observed more details of one event details and observed process metadata for filezilla.exe, Version, Executable path, PID 4688, parent PID 4416, user details, Auth ID, and start time loaded Modules.

Event Properties																																																														
Event																																																														
Image	FileZilla FTP Client																																																													
	FileZilla Project																																																													
Name:	filezilla.exe																																																													
Version:	3.69.3.0																																																													
Path:	C:\Program Files\FileZilla FTP Client\filezilla.exe																																																													
Command Line:	"C:\Program Files\FileZilla FTP Client\filezilla.exe"																																																													
PID:	4688	Architecture:	64-bit																																																											
Parent PID:	4416	Virtualized:	False																																																											
Session ID:	1	Integrity:	Medium																																																											
User:	WINDOWS11\vbouser																																																													
Auth ID:	00000000:00050d77																																																													
Started:	9/25/2025 12:08:43 AM		Ended:	(Running)																																																										
Modules:	<table border="1"> <thead> <tr> <th>Module</th> <th>Address</th> <th>Size</th> <th>Path</th> <th>Company</th> <th>Version</th> <th>Timestamp</th> </tr> </thead> <tbody> <tr> <td>windows.applicati...</td> <td>0x7f98f200000</td> <td>0xb000</td> <td>C:\Windows\System32\windows.applic...</td> <td>Microsoft Corpor...</td> <td>10.0.26100.1 (Wi...</td> <td>5/3/1961 4:57:19 ...</td> </tr> <tr> <td>OLEACC.dll</td> <td>0x7f98fbc0000</td> <td>0x78000</td> <td>C:\WINDOWS\SYSTEM32\OLEACC.dll</td> <td>Microsoft Corpor...</td> <td>7.2.26100.5074 (...)</td> <td>9/25/2034 6:23:3...</td> </tr> <tr> <td>thumbcache.dll</td> <td>0x7f990140000</td> <td>0x8f000</td> <td>C:\Windows\System32\thumbcache.dll</td> <td>Microsoft Corpor...</td> <td>10.0.26100.1 (Wi...</td> <td>3/9/1938 6:17:17 ...</td> </tr> <tr> <td>COMCTL32.dll</td> <td>0x7f990d00000</td> <td>0x293000</td> <td>C:\WINDOWS\WinSxS\amd64_micros...</td> <td>Microsoft Corpor...</td> <td>6.10 (WinBuild.1...</td> <td>5/2/1976 5:53:17 ...</td> </tr> <tr> <td>WINMM.dll</td> <td>0x7f991300000</td> <td>0x35000</td> <td>C:\WINDOWS\SYSTEM32\WINMM.dll</td> <td>Microsoft Corpor...</td> <td>10.0.26100.1 (Wi...</td> <td>8/5/1934 5:43:14 ...</td> </tr> <tr> <td>msimg32.dll</td> <td>0x7f998970000</td> <td>0x80000</td> <td>C:\WINDOWS\SYSTEM32\msimg32.dll</td> <td>Microsoft Corpor...</td> <td>10.0.26100.1882</td> <td>2/11/1958 7:50:1...</td> </tr> <tr> <td>NETAPI32.dll</td> <td>0x7f9998b0000</td> <td>0x1a000</td> <td>C:\WINDOWS\SYSTEM32\NETAPI32.dll</td> <td>Microsoft Corpor...</td> <td>10.0.26100.1150</td> <td>1/28/2035 10:09:...</td> </tr> </tbody> </table>						Module	Address	Size	Path	Company	Version	Timestamp	windows.applicati...	0x7f98f200000	0xb000	C:\Windows\System32\windows.applic...	Microsoft Corpor...	10.0.26100.1 (Wi...	5/3/1961 4:57:19 ...	OLEACC.dll	0x7f98fbc0000	0x78000	C:\WINDOWS\SYSTEM32\OLEACC.dll	Microsoft Corpor...	7.2.26100.5074 (...)	9/25/2034 6:23:3...	thumbcache.dll	0x7f990140000	0x8f000	C:\Windows\System32\thumbcache.dll	Microsoft Corpor...	10.0.26100.1 (Wi...	3/9/1938 6:17:17 ...	COMCTL32.dll	0x7f990d00000	0x293000	C:\WINDOWS\WinSxS\amd64_micros...	Microsoft Corpor...	6.10 (WinBuild.1...	5/2/1976 5:53:17 ...	WINMM.dll	0x7f991300000	0x35000	C:\WINDOWS\SYSTEM32\WINMM.dll	Microsoft Corpor...	10.0.26100.1 (Wi...	8/5/1934 5:43:14 ...	msimg32.dll	0x7f998970000	0x80000	C:\WINDOWS\SYSTEM32\msimg32.dll	Microsoft Corpor...	10.0.26100.1882	2/11/1958 7:50:1...	NETAPI32.dll	0x7f9998b0000	0x1a000	C:\WINDOWS\SYSTEM32\NETAPI32.dll	Microsoft Corpor...	10.0.26100.1150	1/28/2035 10:09:...
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NETAPI32.dll	0x7f9998b0000	0x1a000	C:\WINDOWS\SYSTEM32\NETAPI32.dll	Microsoft Corpor...	10.0.26100.1150	1/28/2035 10:09:...																																																								

4. Used Tool > Stack Summary for captured events of filezilla.exe with kernel networking functions by showing stack summary when performing TCP operations.

Stack Summary

Stack traces during trace:

Name	Count	% Count	Time	% Time	Location	Module	Offset	Path
U <All>	47	100.000000%	0.0000000	0.0000000	n/a <All>	<All>	0x0	<All>
U filezilla.exe	47	100.000000%	0.0000000	0.0000000	n/a filezilla.exe	filezilla.exe(4688)	0x0	C:\Program Files\FileZilla FTP Client\filezilla.exe(4688)
K ntoskrnl.exe + 0x6b3055	17	36.17021%	0.0000000	0.0000000	n/a ntoskrnl.exe + 0x6...	ntoskrnl.exe	0x6b3055	C:\WINDOWS\system32\ntoskrnl.exe
K ntoskrnl.exe + 0x6a1be4	10	21.27660%	0.0000000	0.0000000	n/a ntoskrnl.exe + 0x6...	ntoskrnl.exe	0x6a1be4	C:\WINDOWS\system32\ntoskrnl.exe
K ntoskrnl.exe + 0x6a0b24	3	6.38298%	0.0000000	0.0000000	n/a ntoskrnl.exe + 0x6...	ntoskrnl.exe	0x6a0b24	C:\WINDOWS\system32\ntoskrnl.exe
U ntdll.dll + 0x8d9c	1	2.12766%	0.0000000	0.0000000	n/a ntdll.dll + 0x8d9c	ntdll.dll	0x8d9c	C:\WINDOWS\SYSTEM32\ntdll.dll
K ntoskrnl.exe + 0x6a09fe	1	2.12766%	0.0000000	0.0000000	n/a ntoskrnl.exe + 0x6...	ntoskrnl.exe	0x6a09fe	C:\WINDOWS\system32\ntoskrnl.exe

Filter...

Go to Event Source... Close

5. Used Tool > Network Summary to observe events and their paths which shows 10.0.2.15:29667, 10.0.2.15:46032, and 10.0.2.15:ftp.

Network Summary

Network paths accessed during trace:

Network Time	Total Events	Connects	Disconne...	Sends	Receives	Send Byt...	Receive ...	Other	Path
0.0000000	3	1	1	1	0	26	0	0	10.0.2.15:29667
0.0000000	4	1	1	0	1	0	146	1	10.0.2.15:46032
0.0000000	40	1	0	11	14	108	456	14	10.0.2.15:ftp
0.0000000	47	3	2	12	15	134	602	15	<Total>

Filter... 4 items

Save... Close

Conclusion

This lab helps to cover the practical aspects of the packet analysis and investigate forensic evidence and other details using Wireshark, Network Miner, and Process Monitor. The analysis confirmed that the attacker IP 10.0.2.15 acted as an FTP server (vsFTPD 3.0.5) and the victim host 10.0.2.4 authenticated using captured credentials (username - jane and Password - pwd2345). The details provide file transfer completion from the victim executed STOR *ts.txt* and 226 Transfer complete. Network Miner reconstructed *ts.txt* and flagged content metadata; sessions show FTP data channels on ephemeral ports and an outbound TLS session from the victim 10.0.2.4 to 20.59.87.225 through port 443. Network summaries within ProcMon reflected matching ports 21, 29667, 46032 and TCP event patterns identical to Wireshark logs.

Recommended to use encrypted protocols which cannot intercept credentials and data, deploy real-time endpoint detection and response (EDR) and policy to adopt them, network segmentation and firewalls, use data loss prevention (DLP) controls, and finally provide user security awareness programs periodically to mitigate this kind of incident happened in future.

This lab highlighted the risk of using unencrypted FTP protocol and how the attackers can easily grab user credentials and transfer data using FTP protocol. The tools we used for this lab demonstrate digital forensic methodology by validating attacker actions and attributing network events to specific user processes.

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